

EFY Note

The source code of this project is included in this month's EFY DVD and is also available for free download at source.efymag.com

Construction and testing

The below-mentioned settings must be made in the oscilloscope for proper display of images (any deviation in values will lead to distorted images):

1. Volt/div knob set to 50mV
2. Time/div knob set to 10ms
3. Trigger source set to internal trigger
4. Trigger mode set to auto
5. Trigger slope set to negative (may be neglected)

Now resize the image to 84×32 (width×height) size. Convert the image to monochrome BMP format by saving the resized image as monochrome BMP format in MS-Paint.

Download microC PRO for AVR from http://www.mikroc.com/downloads/get/299/mikroc_pro_avr_v600.zip. Install the software and run it. Open GLCD bitmap editor from the tool menu as shown in Fig. 3.

Get the byte code of the BMP image by directly loading the image in the Nokia 3310 tab as shown in Fig. 4.

Copy the 504 bytes of byte code of the image and paste it in the source code as shown in Fig. 5.

Compile the code and burn it in the microcontroller using a suitable programmer. Connect the oscilloscope probe to the circuit as shown in Fig. 2 and the stored image will be displayed. The code is optimised and properly synchronised with time for proper display of images on the oscilloscope. Any changes in the code (addition of time delays, lines), clock frequency, crystal value, changes in settings of oscilloscope may lead to a distorted image. The code is written so that no external trigger signal is required. The image is displayed with proper synchronisation with the internal trigger signal. ●

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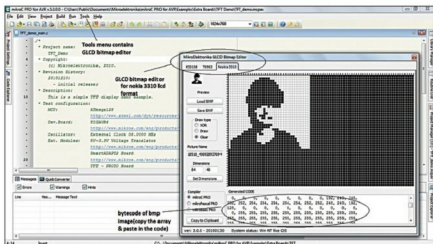


Fig. 3: GLCD bitmap editor

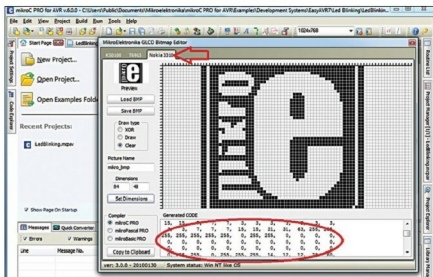


Fig. 4: Loading the BMP image to get the byte code

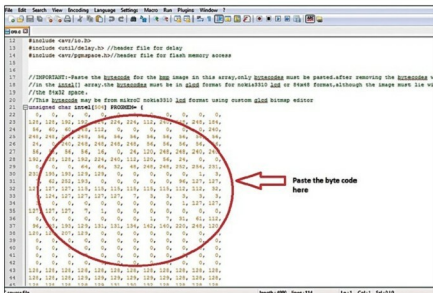


Fig. 5: Copying byte code of the image in the program